

SEATTLE PAINE, WA, USA

WMO: 727937

Lat: 47.908N

Lon: 122.280W

Elev: 185

StdP: 99.13

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 24222

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-3.7	-1.6	-11.2	1.5	-1.0	-8.0	1.9	0.3	12.9	9.5	11.6	8.7	2.2	120	0.539

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.1	27.2	17.7	25.1	16.9	23.2	16.2	18.5	25.5	17.6	23.9	16.8	22.4	2.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.9	11.6	20.2	14.9	10.8	19.6	14.1	10.3	18.8	52.5	25.4	49.8	23.9	47.4	22.3	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.9	9.0	7.8	-6.1	30.8	2.7	2.3	-8.0	32.5	-9.6	33.8	-11.1	35.1	-13.1	36.8		
			-7.2	20.0	2.6	0.9	-9.1	20.7	-10.6	21.2	-12.0	21.7	-13.9	22.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	5.2	5.5	7.3	9.4	12.5	14.9	17.5	17.9	15.4	11.0	7.2	4.6
	DBStd	5.45	3.23	2.96	2.75	2.68	2.84	2.79	2.54	2.21	2.47	2.48	3.21	3.16
	HDD10.0	697	152	126	92	41	8	1	0	0	0	17	93	169
	HDD18.3	2827	408	358	342	267	182	109	46	34	93	226	334	427
	CDD10.0	969	2	1	8	24	85	148	232	246	163	49	9	1
	CDD18.3	56	0	0	0	0	2	7	20	22	6	0	0	0
	CDH23.3	393	0	0	1	2	13	51	154	139	33	0	0	0
	CDH26.7	73	0	0	0	0	2	7	37	24	3	0	0	0
Wind	WSAvg	3.2	3.7	3.4	3.9	3.4	2.9	2.7	2.6	2.5	2.5	3.0	3.6	3.6
Precipitation	PrecAvg	895	114	77	100	60	56	44	16	24	42	88	138	135
	PrecMax	1111	226	213	175	116	90	88	35	72	99	184	270	237
	PrecMin	669	57	34	53	12	12	9	0	0	9	19	61	46
	PrecStd	136	42	40	36	26	23	24	11	21	24	43	58	55
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.8	14.1	18.1	22.4	25.0	27.7	30.2	29.4	26.6	20.2	16.6	12.9
		MCWB	10.9	7.9	10.9	13.8	15.6	17.9	19.5	18.4	16.8	14.1	12.5	10.6
	2%	DB	12.1	12.2	14.9	17.9	21.6	24.8	27.2	26.8	23.9	17.7	14.0	11.1
		MCWB	9.9	8.0	9.4	11.5	14.6	16.7	18.0	17.8	16.0	13.0	11.5	9.1
	5%	DB	10.7	10.9	12.7	16.0	19.3	22.2	24.7	24.7	21.8	16.2	12.6	9.7
		MCWB	8.4	7.7	8.6	10.5	13.6	15.4	17.0	17.0	15.3	12.5	10.5	7.7
	10%	DB	9.4	9.5	11.3	13.7	17.3	20.0	22.7	22.8	19.9	14.7	11.5	8.7
		MCWB	7.4	7.1	8.2	9.4	12.6	14.4	16.2	16.3	14.7	11.8	9.5	6.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.3	10.4	12.0	14.1	16.6	18.9	20.2	19.2	18.1	15.7	13.7	11.8
		MCDB	13.0	11.8	15.7	20.7	23.0	26.5	28.8	26.8	24.3	18.4	15.2	12.7
	2%	WB	10.4	9.3	10.5	12.3	15.1	17.0	18.5	18.3	16.9	14.2	12.1	9.5
		MCDB	11.5	11.1	13.6	17.3	20.8	23.5	25.7	25.2	22.0	16.1	13.6	10.4
	5%	WB	8.9	8.1	9.6	11.0	14.1	15.8	17.4	17.5	15.9	13.1	11.0	8.0
		MCDB	10.0	9.8	12.2	14.6	18.5	21.2	24.1	23.7	20.3	15.2	12.4	9.0
	10%	WB	7.8	7.5	8.5	10.0	12.9	14.9	16.6	16.8	15.2	12.2	9.9	7.3
		MCDB	8.9	9.0	10.9	13.1	16.8	19.4	22.3	22.1	18.9	14.3	11.3	8.3
Mean Daily Temperature Range	5% DB	MDBR	4.4	5.2	6.1	6.9	7.5	7.8	9.2	9.1	7.6	5.8	4.8	4.2
		MCDBR	5.6	7.0	8.3	10.2	10.9	11.4	12.5	12.0	10.9	7.8	5.8	5.2
		MCWBR	4.0	4.4	4.7	5.4	5.7	5.7	5.4	5.1	4.8	4.4	3.9	3.9
	5% WB	MCDBR	4.9	5.4	7.2	9.0	9.7	10.3	11.6	10.8	9.1	6.3	5.1	4.5
		MCWBR	4.1	4.0	4.5	5.2	5.5	5.5	5.5	5.0	4.4	4.3	4.0	3.9
Clear-Sky Solar Irradiance	taub	0.313	0.305	0.316	0.343	0.361	0.349	0.340	0.345	0.333	0.320	0.309	0.314	
	taud	2.527	2.565	2.558	2.461	2.428	2.481	2.530	2.523	2.548	2.584	2.565	2.461	
	Ebn at Noon	773	870	913	911	902	912	917	898	880	839	774	714	
	Edh at Noon	62	72	85	103	110	105	99	96	85	70	59	58	
All-Sky Solar Radiation	RadAvg	0.99	1.87	2.84	4.23	5.34	5.86	6.53	5.54	4.00	2.24	1.16	0.80	
	RadStd	0.14	0.24	0.33	0.39	0.43	0.56	0.48	0.36	0.34	0.25	0.16	0.11	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.48	-1.10	-2.22	+1.48	N/A	N/A	N/A	-159	+157	+33	
(47)	Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page